

Architectural Addendum: Stratified Batching & Asymmetric Temporal Clocks

Context: This addendum expands upon the core principles of the Inverted Model-Agnostic Meta-Learning (MAML) hardware architecture. It formally describes the methodology for acquiring and integrating differential weight updates across a continuously degrading analog system, specifically addressing the physics of charge leakage in Analog In-Memory Computing (AIMC) matrices.

1. The Problem with Point-in-Time Momentum Updates

In standard machine learning frameworks, weights are updated continuously based on point-in-time gradients (e.g., Stochastic Gradient Descent coupled with Momentum). While effective in digital domains, this approach fails in analog architectures where the "target" is continuously shifting due to physical leakage (voltage drop over an RC-curve).

If the system samples a single differential temporal window and immediately applies an update, the calculated gradient only represents the error at that precise point in the discharge cycle. Using Momentum to smooth these updates across multiple refresh cycles introduces severe *phase lag*. The weights perpetually attempt to compensate for a historical hardware state, often leading to harmonic oscillation and architectural instability.

2. Stratified Batching Over the Discharge Cycle

To eliminate temporal lag and prevent cyclic overcompensation, the system utilizes a **Stratified Batching** methodology. Instead of immediate updates, the discharge cycle between refresh pulses is divided into a discrete number of evenly distributed temporal/voltage regions (e.g., 10 fixed "strata" or slices).

- **Data Acquisition:** During a single full discharge cycle, the digital controller extracts differential gradients at each of the 10 distinct strata.
- **Global Error Minimization:** These gradients are aggregated into a single batch. By averaging the derivatives across the entire RC-discharge curve, the non-linear nature of the physical leakage is mathematically integrated out.

- **Optimal Initialization:** The resulting weight update, applied only at the global refresh pulse, represents the optimal global compromise. The Inverted MAML algorithm thus computes the exact initial weight configuration that minimizes the integral of the residual error across the *entire subsequent discharge cycle*.

[DIAGRAM: DISCHARGE CURVE DIVIDED INTO 10 REGIONS]

Mathematical Perspective

Instead of optimizing W for a specific time t , the controller minimizes the loss L over the sequence of strata $S = \{t_1, t_2, \dots, t_{10}\}$ via $\nabla W = (1/10) \sum \nabla L(t_i)$. This embeds the deterministic trajectory of the leakage into the meta-learned initialization.

3. The Asymmetric Discharge Clock (Symmetry Breaking)

A fundamental challenge in evaluating analog matrices is that *all* weights within the hybrid triad (payload, compensation, and reference) are simultaneously discharging. If all components leaked at an identical rate, the system would become entirely unobservable; the algorithm would have no fixed reference point to determine absolute drift, leading to catastrophic model collapse.

This challenge is resolved by deliberately introducing an **Asymmetric Discharge Rate**. The dedicated Reference cells are engineered to discharge significantly faster than the compensation and payload weights.

- **The Temporal Anchor:** The rapid leakage of the Reference cells provides a full dynamic swing, effectively acting as an intrinsic analog clock.
- **Symmetry Breaking:** Because the reference moves aggressively, it creates a dominant, clean gradient signal during the differential evaluation windows. This strong signal drowns out the micro-noise generated by the slow drift of the compensation weights.
- **Attractor State Formation:** The fast reference forces the MAML algorithm to anchor itself. It optimizes the residual difference between the payload and the correction matrix strictly relative to the predictable, rapid decay of the reference clock.

Summary of Impact

By combining **Stratified Batching** (to learn the geometry of the entire discharge curve) with an **Asymmetric Reference Clock** (to break physical symmetry and provide a strong gradient anchor), the Inverted MAML architecture converts parasitic hardware leakage from a detrimental error source into a predictable computational feature. The system does not fight the physics of the hardware; it meta-learns to synchronize with it.